Viridis Research — certification record

Run `runs/2026-11-12_P4` 2026-11-12 project P4, task P4-T001 engine `2368d59583c5`
GitHub run: `https://github.com/jdhart81/viridis-engine/actions/runs/43134889263`

Do wildfires that start in smaller barrier compartments end smaller? A re-executed pre-registered confirmatory test in northeast Washington
Justin Hart, Viridis LLC

Labels awarded by the verifier

Claim	Label	Claim text (as submitted)
C1	**TESTED**	empirical: Running the frozen registered P1C v1.4 script (code/run_test.py, unchanged, via code/rerun_p1c.py) on the registered input files gives, for the registered primary test (H2, definition B, model M2R): n = 11,314 ignitions, beta = 0.0531 (0.05313557) log10 final fire size per log10 compartment area, max(two-way, year-clustered) SE = 0.0120, registered interval beta +/- 2.53 x max SE = [0.0228, 0.0835], decision SUPPORTED under the frozen rule (SUPPORTED iff beta - C2 x max SE > 0), and pre-specified reading REPLICATED. The script's input check against the registered FREEZE.json passed (inputs_ver

Other checks

- Lean toolchain leanprover/lean4:v4.28.0, Mathlib 8f9d9cff6bd7; kernel replay by `leanchecker`.
- References: 15 entries, 11 DOIs, 11 resolved, 0 unresolved, 0 unchecked.
- Agent paper: compiled.
- Statement proposals: 0.

How to read this record

ASSUMED: not verified (the default). **PROVEN**: a Lean 4 proof of a statement sealed before the attempt, compiled with the pinned toolchain, matched definitionally to the sealed statement, using only the axioms propext, Classical.choice and Quot.sound, and replayed through the kernel. **TESTED**: the verifier re-ran the claim's analysis script in a sandbox with no network, on input files pinned by hash, and reproduced the cited result files; this does not certify where the data came from or the inference drawn from it. **REPLICATED**: not yet enabled. **ENDORSED/ADOPTED**: granted by people. Novelty is never a label. **TRIVIAL** marks a sealed statement that one of the engine's probe tactics already closes.

This paper was drafted by an AI research agent (Claude, by Anthropic) running the Viridis research engine; formal proof search may use Aristotle (Harmonic). Justin Hart directs the research program. The labels on this page were assigned by an automated verifier, not by the agent, and only they carry certification. Claims in the paper that are not listed here as PROVEN or TESTED are ASSUMED.

Reproduce everything from `REPRODUCE.md` in the run folder; the run's files are listed with hashes in `MANIFEST.sha256`.

Do wildfires that start in smaller barrier compartments end smaller? A re-executed pre-registered confirmatory test in northeast Washington

Justin Hart
Viridis LLC

12 November 2026

Abstract

Trees Cannot Cut holds that wildfires end where closed loops of holdable features stop them. One consequence is that a fire which starts inside a smaller compartment of mapped barriers should end smaller. After a pilot in the Okanogan, Methow and Chelan region of Washington State, a confirmatory replication (P1C v1.4) was pre-registered for the adjoining northeast Washington region. Its plan, code, blind predictor table and critical values were hash-frozen and timestamped before any fire size was requested, and the plan went through five independent audits. The single confirmatory test regresses $\log_{10}$ final fire size (FPA FOD, 1992–2024) on $\log_{10}$ compartment area, where compartments are bounded by TIGER roads, NHD water and LANDFIRE non-burnable fuel. It controls for the gridMET energy-release-component percentile, fuel group, cause, month, year, barrier density, road distance and 40 km regional fixed effects (model M2R), and it is decided on the interval $\beta \pm 2.53 \times \max(\mathrm{SE}_{\text{two-way}}, \mathrm{SE}_{\text{year}})$. We re-ran the frozen registered script, unchanged, on the registered input files, and every file's SHA-256 hash matched the registration. The re-run gives the published numbers. On 11,314 ignitions, $\beta = 0.0531$ (registered interval 0.0228 to 0.0835, $T = 4.42$), so the decision under the frozen rule is SUPPORTED and the pre-specified reading is REPLICATED. In plain terms, a compartment ten times larger goes with final fires about 13% larger. The study's own pre-specified checks set firm limits on this result. **The study is observational. It does not show that barriers stop fires**, and it cannot say how much of the association runs through suppression access. **Without the 40 km regional terms** (S13, the pilot's model M2) the estimate is 0.033, and it does not pass its critical value of 3.93. **Lightning-caused fires show no effect** (S6a: −0.015, 95% CI −0.062 to 0.033); the association is in human-caused fires. **Counting burn scars from the previous ten years as barriers** shrinks the estimate to 0.020 (S5).

1 Introduction

Fire managers rely on roads, water bodies and ground that will not burn as places where a fire can be held. *Trees Cannot Cut* [4] argues that the structure of these features matters as a network: a fire ends where a closed loop of holdable features stops it, so the size of the loop that encloses an ignition (its *compartment*) should bound how large the fire can grow. If that holds, a fire that starts in a small compartment should, on average, end smaller than one that starts in a large compartment, once fire weather, fuel and access are held equal. The question matters for landscape design. If compartment size predicts fire size, the arrangement of barriers, and not only their total amount, could be managed to limit fire spread. It also connects to operational work on potential control locations and pre-planned containment [7, 8].

This question was put to wildfire records in October 2026 in two steps. A pilot in the Okanogan, Methow and Chelan region (2,677 fires) found the association for mapped barriers. A confirmatory replication, P1C, was then designed for the adjoining northeast Washington region. It was frozen and registered before any fire size from that region was requested (viridis-canon pull request 138; 5), and was run once on 22 October 2026.

This paper reports that registered study in full and adds one element: an independent re-execution. We re-ran the frozen script, unchanged, on the registered input files, and report whether it gives the published numbers. It does. We add no analysis, subgroup or model. Every number below is a pre-specified output of the frozen script.

2 Background and related work

The pilot. The pilot used FPA FOD 1992–2024 for the Okanogan, Methow and Chelan region, frozen before any fire size was fetched. Its decisive test (H2) asked whether compartment size predicts final fire size beyond barrier density and road distance. Under definition B (mapped barriers) it estimated $\beta = 0.063$ (95% CI 0.010 to 0.117, $n = 2{,}677$), about 16% larger fires per tenfold larger compartment. A second definition, A50, used the Washington DNR potential control location (PCL) raster [8] thresholded at 0.5. It was underpowered and was not supported.

Why a confirmatory test. The pilot's estimate cleared $p < 0.05$, so it is probably inflated (winner's curse). The P1C plan also re-read the pilot under alternative spatial error models and found its evidence only moderate. The confirmatory test therefore needed new data, a single pre-specified test, and a critical value calibrated for spatial correlation. The plan uses the small-telescopes benchmark of Simonsohn [13] to define what a failed replication would mean.

Data sources. Ignitions and final fire sizes come from the Fire Program Analysis fire-occurrence database (FPA FOD), 7th edition, FPA_FOD_20260615 [12], built as described by Short [11]. Barriers come from TIGER/Line roads, vehicle classes only [14]; NHD large-scale water areas and waterbodies [15]; and the non-burnable classes of LANDFIRE 2024 FBFM40 [6, 9, 10]. Fire danger is the gridMET energy release component, expressed as a per-cell percentile over 1992–2024 [1]. Burn scars for sensitivity S5 are MTBS perimeters, used as geometry only [3]. Control locations for the secondary definitions A50 and A25 come from the DNR PCL raster [8].

3 The pre-registration

3.1 Freeze, registration and outcome fetch

The plan, code, blind predictor table, raw inputs, logs and critical-value records were hash-frozen in `FREEZE.json` and submitted to OpenTimestamps. Outcomes (fire size) were requested only after three guards passed: G1, every frozen file unchanged; G2, an anonymous download of the public plan and `FREEZE.json` byte-identical to the local copies; G3, no shared ignition with the pilot. The published report gives this order of events on 22 October 2026 (UTC): registration record written at 11:30:09; addendum registration at 11:30:17; the guard download at 11:30:56; all 14,105 of 14,105 outcomes fetched by 11:34:43, none missing and none overlapping the pilot; the fetch record timestamped at 11:34:46. `run_test.py` then ran once. The report records no deviations from the plan beyond Addendum 1, which changed no frozen file, model, statistic, critical value or decision rule.

3.2 Five audits

The plan passed through five versions, each frozen, timestamped and audited by a separate agent that had not built the study. None was published before v1.4.

1. **Audit of v1.0** found an inaccurate blindness statement about MTBS attributes, a registration guard that trusted a local file, neglect of the 33 years shared with the pilot, unfrozen fetch evidence and undisclosed post-fire map updates. v1.1 fixed these and also replaced the pilot's interval with $t_{0.975,32} \times \max \mathrm{SE}$.

2. **Audit of v1.1** found that this calibration held only if spatial correlation stops at the 20 km block edges. Under correlation that crosses block edges, the rule gave 6–8% false calls. v1.2 calibrated a critical value $C_2 = 2.31$ under four spatial error models fitted to the pilot's residuals.

3. **Audit of v1.2** found that the fitted range of residual correlation (11 km) is biased low. Under fields with ranges of 22–45 km, the rule gave 6.7–9.3% false calls. v1.3 calibrated over a screened family of error models that the pilot's residuals cannot reject (140 exponential fields and 9 block models), and changed the primary model to M2R, which adds 40 km regional fixed effects. That choice was recorded before the pilot's M2R estimate was computed.

4. **Audit of v1.3** found that $C_2 = 2.37$ had no margin against anisotropic fields and compartment-level clustering, both outside the family, and that S13's critical value undercovered exact long-range fields. v1.4 added a stress set of 90 structures fixed in advance, of which 52 are retained. This raised the critical values to $C_2 = 2.53$, $C_1 = 2.08$ and $C_{2,\mathrm{M2}} = 3.93$.

5. **Audit of v1.4** (a verification audit) found no critical or moderate issue and six minor ones. Addendum 1, written and timestamped before any outcome was requested, answered them. It noted that between grid points a retained model could set a critical value up to about 0.08 higher for C_2 (2.61) and about 0.16 higher for $C_{2,\mathrm{M2}}$ (4.10). It therefore labels as *borderline* any verdict with $2.53 < T \leq 2.61$ (primary) or $3.93 < T \leq 4.10$ (S13). These notes change no decision.

Up to v1.2 every change widened the interval. v1.3 changed the model, and v1.4 changed only the critical values, which can only rise. All changes were made before any outcome from the region existed.

4 Region, data and sample

The analysis box is longitude −119.4 to −117.0 and latitude 47.75 to 49.0. It covers eastern Okanogan, Ferry, Stevens and Pend Oreille counties, the northern edges of Lincoln and Spokane counties, and a strip about 3 km wide of the Idaho panhandle. The region adjoins the pilot to the east. Predictors were computed on a 30 m grid in EPSG:5070 over the analysis box plus 15 km. The sample is every FPA FOD record inside the box: 14,105 ignitions (4,779 natural, 9,080 human, 246 missing cause), none shared with the pilot. After the predictor-based rules (ignitions on holdable cells and censored compartments excluded), 11,314 ignitions are usable under definition B.

Definitions. **B** (primary): compartments bounded by roads, water and non-burnable ground. **A50**/**A25**: PCL > 0.5 / > 0.25. **B+**: B plus MTBS scars from the previous 10 years. The PCL raster has large no-data gaps in this region, so A50 and A25 samples are small (102 and 712).

Known limits of blindness (disclosed in the plan). During the pilot, MTBS perimeters with full attributes were received for the pilot's compute box, and 63 of them reach into this region. Only geometry and year were kept. Sensitivities S10 and S10b drop the exposed ignitions. The fuel and road maps date from 2024 but are applied to fires from 1992 to 2024. LANDFIRE updates fuels after disturbance (addressed by S11), and post-fire roads would work against the hypothesis. Five ignitions have a pilot ignition on the same date within 5 km; S12 drops them.

5 Model and decision rule

The outcome is $Y = \log_{10}(\text{fire size, acres})$. The models are

$$\begin{aligned} \text{M1:} &\quad Y \sim \log_{10} A_{\text{enc}} + \text{ERC}_{\text{pct}} + \text{fuel group} + \text{cause} + \text{month} + \text{year}, \\ \text{M2:} &\quad \text{M1} + \log_{10}(\rho + 0.001) + \log_{10}(\text{road distance} + 30\,\text{m}), \\ \text{M2R:} &\quad \text{M2} + 40\,\text{km region}, \end{aligned}$$

with categorical terms as fixed effects, estimated by OLS. A_{enc} is the area of the compartment that encloses the ignition and ρ is barrier density. The 40 km regions are the cells $(\lfloor x/40000 \rfloor, \lfloor y/40000 \rfloor)$ in EPSG:5070, each the 2×2 parent of the 20 km clustering blocks.

The single confirmatory test is H2, definition B, model M2R: $\beta(\log_{10} A_{\text{enc}}) > 0$. Let SE_2 be the CR1 standard error two-way clustered by year and 20 km block [2], and SE_{year} the year-clustered CR1 standard error. With $T = \beta / \max(\text{SE}_2, \text{SE}_{\text{year}})$, the registered interval is $\beta \pm C_2 \max(\text{SE}_2, \text{SE}_{\text{year}})$, $C_2 = 2.53$. The verdict is SUPPORTED if the interval lies above 0, CONTRADICTED if it lies below 0, and NOT SUPPORTED otherwise. The pre-specified reading is REPLICATED if SUPPORTED. It is NOT REPLICATED if NOT SUPPORTED and the one-sided bound $\beta + C_1 \max \text{SE}$ ($C_1 = 2.08$) falls below $\beta_{33} = 0.0415$, the effect the pilot had 33% power to detect [13]. It is INCONCLUSIVE otherwise. The frozen decision code is:

```
def decide_ci(lo, hi):
    return 'SUPPORTED' if lo > 0 else ('CONTRADICTED' if hi < 0 else 'NOT␣SUPPORTED')

def decide_primary(tw):
    """The registered P1C decision: the interval beta +- C2 * max(two-way SE, year-only
        SE)."""
    if tw.get('not_estimable'):
        return 'NOT␣ESTIMABLE'
    return decide_ci(tw['interval_registered'][0], tw['interval_registered'][1])
```

Across the error models in the plan's power table, the simulated probability of SUPPORTED was 0.52–0.60 at β_{33} and 0.86–0.95 at the pilot's M2 estimate (0.063).

6 Re-execution

We downloaded the eleven registered input files and the frozen `run_test.py` from the registration repository at commit `cfdf55a24b2b833ef88dcd160813edab9c7116dd`, gunzipped the three compressed tables, and checked each SHA-256 hash against the registered list. All twelve matched. The entry script `rerun_p1c.py` copies the inputs into the folder layout the frozen script expects and runs it unchanged with `python3 -B`. It then writes the full output (`p1c_results.json`) and the primary test (`p1c_primary.json`). The frozen script itself refuses to run unless every input

matches `FREEZE.json` and the outcomes match the fetch record; it reported `inputs_verified: true`.

We ran it with Python 3.12.3, numpy 2.5.4, pandas 3.1.1 and scipy 1.18.1 (the frozen script imports `scipy.stats` for a t critical value). Its only randomness, the 2,000-replicate year bootstrap and the 10,000 H4 permutations, uses a fixed generator seed (20261020). Two runs from an empty results folder gave byte-identical files. Every number reported below agrees with the published report to the precision it gives. The SHA-256 of our `p1c_primary.json` differs from the hash recorded on 9 November 2026 with numpy 2.5.3 and pandas 3.0.5. To see why, we also ran the package in a scratch copy with numpy 2.5.3 and pandas 3.0.5 on our machine. That gave a third hash. Across both runs, all decisions, readings and counts were identical, and the largest relative difference in any number was 9×10^{-13} in `p1c_primary.json` and 6×10^{-12} in `p1c_results.json`. The hashes therefore differ only through floating-point rounding in the last digits, which depends on the platform and linear-algebra libraries; see Section 10.

7 Results

7.1 Primary result

Table 1: The registered primary test (H2, definition B, model M2R), from `results/p1c_primary.json`.

Ignitions in the primary sample	11,314
Year clusters / 20 km blocks / 40 km regions	33 / 75 / 24
β, $\log_{10}$ size per $\log_{10}$ compartment area	**0.0531**
SE_2 / SE_{year} / max	0.0120 / 0.0097 / 0.0120
Registered interval, $\beta \pm 2.53 \times \max SE$	**0.0228 to 0.0835**
$T = \beta / \max SE$	4.42
Decision (frozen rule)	**SUPPORTED**
Pre-specified reading	**REPLICATED**
Fire size per tenfold larger compartment	$\times 1.130$

$T = 4.42$ lies outside the borderline band $(2.53, 2.61]$ of Addendum 1. Other readings are reported but do not decide. The v1.1-rule interval ($t_{0.975,32} \times \max SE$) is 0.0287 to 0.0776, which is SUPPORTED, and the pilot-rule reading is also SUPPORTED. The year-clustered 95% CI is 0.0342 to 0.0721 and the year cluster bootstrap gives 0.0350 to 0.0721. For the small-telescopes reading, $\beta_{33} = 0.0415$ and the 90% upper bound is 0.078, so effects as large as those the pilot could detect are not rejected. The barrier-density and road-distance coefficients are -0.049 and 0.141.

7.2 Pre-specified checks

S13, the pilot's model M2 without regional terms ($C_{2,M2} = 3.93$; no decision weight). $\beta = 0.0330$, $\max SE = 0.0142$, $T = 2.33$. The registered interval is -0.023 to 0.089, so the verdict is NOT SUPPORTED against 3.93, and also against 4.10, so it is not borderline. Its year-clustered 95% CI is 0.005 to 0.061 and its v1.1-rule reading is SUPPORTED. *The estimate is smaller without the 40 km regional terms.* The plan notes that M2 is exposed to differences between reporting agencies. It also notes that M2R compares compartments only within regions, so an effect that worked only between regions would not be detected by the primary test.

Leave-one-year-out. β ranges from 0.049 to 0.057 over the 33 years, and no year's removal changes the decision.

Comparison with the pilot (M2R). The pilot's M2R estimate is 0.079, so the difference is -0.026 ($z = -0.98$, $p = 0.33$ with two-way SEs, from the stacked model). The pooled GLS estimate (two-way) is 0.057 (0.036 to 0.079).

Table 2: Sensitivity analyses, definition B. 95% CIs; the registered-rule interval is given where the plan specifies one.

Check	Model	n	β (95% CI)	Registered rule
S10 outside the pilot's compute box	M2R	9,941	0.043 (0.016–0.069)	SUPPORTED (0.009–0.077)
S10b also excl. pilot-fetched MTBS	M2R	9,368	0.042 (0.012–0.072)	SUPPORTED (0.003–0.081)
S11 no fuel-group term	M2R	11,314	0.051 (0.027–0.075)	SUPPORTED (0.020–0.082)
S12 without possible duplicates	M2R	11,309	0.052 (0.029–0.076)	SUPPORTED (0.022–0.083)
S1 snap ignitions on holdable cells	M2	13,707	0.057 (0.039–0.075)	n/a
S2 include censored compartments	M2	11,585	0.031 (0.012–0.050)	n/a
S3 drop coarse coordinates	M2	11,297	0.033 (0.014–0.052)	n/a
S5 B+ (prior 10-yr burn scars)	M2	11,105	**0.020** (0.001–0.039)	n/a
S6a lightning-caused only	M2	4,226	**−0.015** (−0.062–0.033)	n/a
S6b human-caused only	M2	6,915	0.054 (0.031–0.076)	n/a
S8 years 2010 on	M2	4,865	0.049 (0.025–0.073)	n/a

Secondary models (not confirmatory). B under M1: 0.063 (0.042–0.085), $n = 11{,}314$. A50 under M1: 0.553 (0.361–0.746) and under M2: 0.527 (0.366–0.689), both $n = 102$. A25 under M1: 0.158 (0.087–0.228) and under M2: 0.140 (0.071–0.209), both $n = 712$. Because of the PCL gaps the A50 sample is small. The PCL raster is a model product that may have been trained on fire perimeters from this region.

H3, escaped fires (B, M2, fires ≥ 10 acres, $n = 802$): $\beta = 0.091$ (0.001–0.181).

H4, containment (B, $n = 802$): the observed share of fires no larger than their compartment is 0.946, against a permutation null mean of 0.929 (within year $\times$ fuel-group strata), one-sided $p = 0.0002$.

8 Discussion and limitations

Under the rule frozen and timestamped before any outcome was requested, the compartment association found in the pilot replicates in the adjoining region. It is slightly smaller than the pilot's estimate under the same model (0.053 against 0.079), and the two estimates are statistically compatible. In plain terms, after fire weather, fuel, cause, month, year, barrier density, road distance and 40 km region are taken into account, fires that start in a compartment ten times larger end about 13% larger.

The study's own design and pre-specified checks limit what this shows:

- **Observational.** The study does not show that barriers stop fires. Nor can it separate fuel discontinuity from suppression access: roads that bound small compartments also bring crews and engines. Road distance and barrier density are controlled, but not experimentally.

- **S13: without the regional terms the estimate is 0.033 and does not pass its critical value** (3.93). The result therefore depends on comparing compartments within 40 km regions. This model was chosen, before any outcome existed, for robustness to long-range spatial correlation.
- **S6a: lightning-caused fires show no effect** (-0.015, CI -0.062 to 0.033). The association is in human-caused fires (S6b, 0.054). Human ignitions cluster near roads and settlements, where access and early detection also differ, so this pattern fits a role for suppression access but does not identify one.
- **S5: counting recent burn scars as barriers shrinks the estimate to 0.020.** S2 (censored compartments included, 0.031) and S3 (coarse coordinates dropped, 0.033) also give smaller estimates than the primary.
- **Residual statistical risk.** The critical values cover the screened family and the stress set. They do not cover structures outside both, such as correlation that changes from year to year, combinations of structures at their limits, or non-exponential shapes. Between grid points the binding value could be up to about 0.08 higher. The primary $T = 4.42$ is far from either bound.
- **Maps and blindness.** The barrier and fuel maps are from 2024 and are applied to fires from 1992 onward. The disclosed MTBS exposure is addressed by S10 and S10b, which do not change the decision.
- **Scope of the re-execution.** Our re-run shows that the frozen code gives the published numbers from the registered files. It does not check those files against the agencies' servers, the registration's timestamps, or the blindness of the original fetch. The frozen script checks the inputs against `FREEZE.json` and trusts the fetch record. The study has not been peer reviewed.

9 Conclusions

The pre-registered P1C confirmatory test, re-executed unchanged on the registered inputs, gives $\beta = 0.0531$ (registered interval 0.0228 to 0.0835, $n = 11{,}314$). The decision is SUPPORTED and the reading is REPLICATED: fires that start in a tenfold larger compartment of mapped barriers end about 13% larger. This is an observational association and does not show that barriers stop fires. It weakens to 0.033, below its critical value, without the 40 km regional terms (S13). It is absent in lightning-caused fires (S6a), and it shrinks to 0.020 when recent burn scars count as barriers (S5). Mechanism remains open, in particular whether suppression access rather than fuel discontinuity carries the association.

10 Reproducibility

The package holds `code/run_test.py` (the frozen registered script, SHA-256 `25abae84...`), `code/rerun_p1c.py` (the entry script, `065fe6a5...`), the eleven registered inputs under `data/` with their URLs and hashes in `data_pins.json`, and the outputs `results/p1c_primary.json` and `results/p1c_results.json`. From the package folder, with an empty `results/`, run `python3 code/rerun_p1c.py`; it uses no network and takes a few minutes. It requires numpy, pandas and

scipy. Our run used numpy 2.5.4, pandas 3.1.1 and scipy 1.18.1, and gave `p1c_primary.json` with SHA-256 `6741918a...` and `p1c_results.json` with `adda924a...`.

The entry script is:

```
proc = subprocess.run([sys.executable, "-B", "run_test.py"], cwd=p1c, capture_output=True
    , text=True)
...
if results.get("inputs_verified") is not True or results.get("synthetic") is not False:
    sys.stderr.write("run_test.py␣did␣not␣verify␣its␣inputs\n")
    return 2
primary = {"registered_test": "P1C␣v1.4,␣H2,␣definition␣B,␣model␣M2R",
          "decision_rule": "SUPPORTED␣iff␣beta␣-␣C2␣*␣max(SE_twoway,␣SE_year)␣>␣0", ...}
```

The plan, addendum, audits, logs and published report are in the registration repository [5].

Changelog

- 2026-11-12 (P4-T001): first version. Re-executed the registered P1C v1.4 confirmatory test unchanged on the registered inputs and reported the registered study in full. No new analysis.

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